



RAMCO INSTITUTE OF TECHNOLOGY
Department of Civil Engineering
Academic Year 2020 – 2021



Report of the Minutes of Meeting of PAQIC Meeting - I

Date of Meeting: 05.10.2020 (Monday)

Time of Meeting: 3:30 PM to 5:00 PM

Reference: PAQIC Meeting / 2020

Members Present:

Sl. No.	Name of the Faculty Member	Designation	PAQIC
01.	Mr. S. Dharmar	Assistant Professor (Sr.Gr.) and Head / Civil	Convener
02.	Dr. K. Basarikodi	Professor & Head/ Maths	Member
03.	Mr. G. Karthikeyan	Assistant Professor (Sr.Gr.) / Civil	Member
04.	Mr. R. Muruganantham	Assistant Professor / Civil	Member
05.	Mr. A. Manicka Mamalan	Assistant Professor / Civil	Member
06.	Mrs. V. Anusuya	Assistant Professor (Sr.Gr.) / CSE	Member
07.	Ms. S. Sharmila Kumari	Assistant Professor / EEE	Member
08.	Mrs. V. SreengaNachiyar	Assistant Professor (Sr.Gr.) / ECE	Member
09.	Mrs. C. Subha	Assistant Professor (Sr.Gr.) / Civil	Representative
10.	Mrs. D. Darling Helen Lydia	Assistant Professor / Civil	Representative
11.	Mr. T. Chockalingam	Assistant Professor / Civil	Representative
12.	Mrs. R. Kalaimani	Assistant Professor / Civil	Representative
13.	Mr. J. Ram Prashath	Assistant Professor / Civil	Representative
14.	Mrs. L. Johncy Rani	Assistant Professor / Civil	Representative
15.	Mrs. A. Leema Margret	Assistant Professor / Civil	Representative

1. DISSEMINATION OF VISION, MISSION, PEOs AND PSOs

Presenter: Mr. G. Karthikeyan, Assistant Professor (Sr.Gr.)/Civil

The Vision and Mission, PEOs and PSOs statements of the department and the procedure followed to frame the same were presented to the PAQIC members.

Vision of the Department:

Create knowledgeable and professionally competent Civil Engineers who are capable of providing global solutions to the problems related to Civil Engineering.

Mission of the Department:

- M1: Impart high quality education and technical skills to transform students into Professional Civil Engineers.
- M2: Establish state-of-the-art infrastructure for updating knowledge in recent developments and cutting edge technology.
- M3: Bridge the gap between academics and industry by practical training and providing solutions for design projects related to Civil Engineering.
- M4: Motivate the faculty and students in the teaching learning process for achieving excellence to meet the global standards.
- M5: Produce graduates with high professional and ethical standards for fulfilling the needs of the industry and society.

Programme Educational Objectives:

- Graduates of the programme will be professionally competent and apply the fundamental concepts of mathematics, science and Civil Engineering to solve problems in Civil Engineering.
- Graduates of the programme will perform their responsibilities as professional Civil Engineers with quality and ethics.
- Graduates of the programme, as part of an organization or as Entrepreneurs, will continue to learn with evolving technologies.
- Graduates of the programme will innovate and provide global solutions for a sustainable built environment.
- Graduates of the programme will contribute to the development of sustainable Infrastructure for the betterment of society.

Programme Specific Outcomes:

On completion of the degree, the students will be able to:

- Communicate and present civil engineering projects effectively.
- Use the techniques, skills, and modern engineering tools necessary for civil engineering practice and project management.
- Provide sustainable solutions to civil engineering problems.
- Perform as design consultants in construction industry for the design of civil engineering structures.

Formulation procedure:

- The Vision and Mission of the Department is developed in line with the Vision and Mission of the Institute. The Department of Civil Engineering has established its Vision and Mission through a consultative process involving the stakeholders of the programme including Management, Governing Council members and faculty members.
- Inputs from Internal stake holders and External stake holders, by referring the reputed technical institutes' websites, expectations and needs of students, parents, society and industry are considered for establishing the vision and mission of the department.
- Formulated Vision and Mission statements duly recommended by the Head of the Institution were presented.

Modes of Dissemination:

The member explained the Publishing Modes of Vision, Mission, PEO's and PSO's in the College official Website, HoD Cabin, Notice Board, Class rooms, Faculty Halls, Laboratories, Lab Manuals, Seminar Hall, Students Handbook, Department Newsletters and Hostel Notice Board.



***Fig. 1. Dissemination of Vision, Mission, PEOs AND PSOs, Presented by
Mr.G.Karthikeyan, AP(SG)/Civil***

Remarks from Members:

- The committee members suggested documenting the Vision, Mission, PEOs and PSOs and suggestions received from the internal and external stakeholders.

2. ATTAINMENT OF POs, PSOs AND QUALITY OBJECTIVES

Presenter: Mr.T.Chockalingam, AP/Civil Engineering

- The procedure for calculating CO, PO and PSO attainment for various courses were presented.
- Various assessment techniques such as direct assessment and indirect assessment techniques for theory subjects, laboratories and projects were discussed.
- CO attainments for the Academic Year 2019-2020 was discussed.
- Sample CO-PO and CO-PSO mapping is given below for reference.

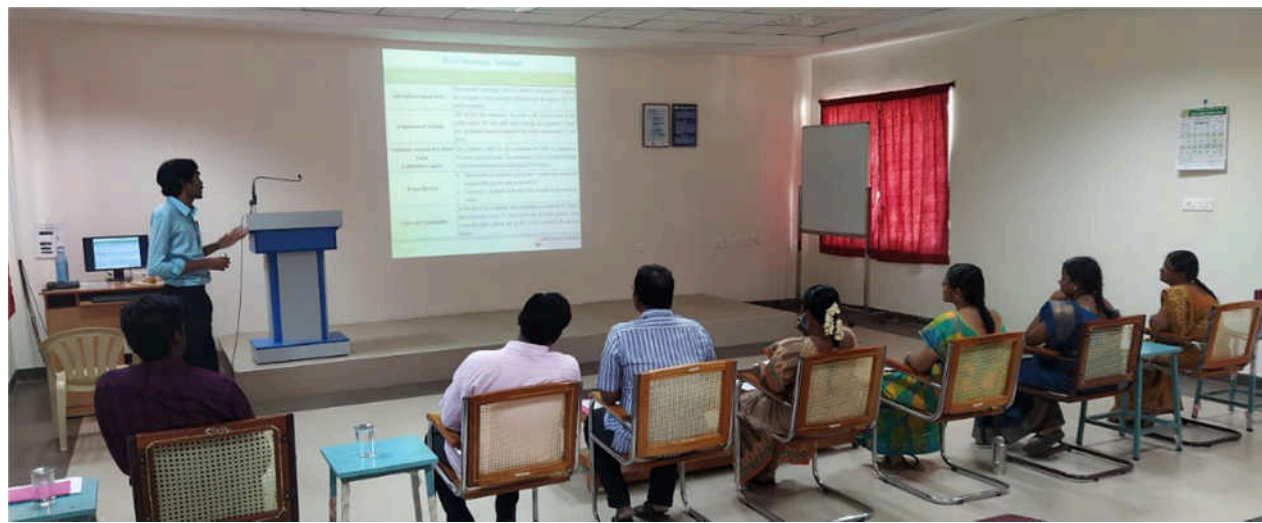
Course Code & Title : CE8301 Strength of Materials I Semester & Year of study : III Semester & 2019 - 2020 Course Index : C211												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C211.1	3	2	2	2	2	1	1	1	-	-	-	-
C211.2	2	3	2	2	2	1	1	1	-	-	-	-
C211.3	2	3	2	2	2	2	1	1	-	-	-	-
C211.4	2	1	3	2	1	1	1	1	-	-	-	-
C211.5	2	3	2	1	2	1	1	1	-	-	-	-
C211	2	2	2	2	2	1	1	1	-	-	-	-

Course Code & Title : CE8301 Strength of Materials I Semester & Year of study : III Semester & 2019 - 2020 Course Index : C211				
CO	PSO1	PSO2	PSO3	PSO4
C211.1	1	2	2	2
C211.2	1	2	2	2
C211.3	1	2	2	2
C211.4	1	2	2	2
C211.5	1	2	2	2
C211	1	2	2	2

- The attainment of CO is calculated by giving 60% weightage to university results and the remaining 40% to Internal Assessment Test.
- The Overall Attainment for the course is calculated as follows:

$$\text{Overall CO Attainment} = \left(\frac{\sum_{i=1}^n CO_i}{n} \times 0.4 \right) + (UA \times 0.6)$$

Where n= number of course outcome.



*Fig. 2. Attainment of PO's, PSO's and Quality Objectives, Presented by
Mr. T. Chockalingam, AP/Civil*

Remarks from Members:

- Unattained COs are indicated and discussed. The root cause analysis for not attaining the target is to be carried out and the same should be discussed with faculty members.
- New pedagogy techniques should be implemented to attain the target as specified.
- The committee members suggested comparing the attainment obtained for the academic year 2019-2020 with previous year's attainment.

3. PROGRAM EFFECTIVENESS

Presenter: Mrs.C.Subha, AP (S.G)/Civil

Project thrust areas:

Students are encouraged to carry out their project in various diversified thrust areas which results in Material Quality Analysis, Replacement of conventional construction materials by low cost eco-friendly materials, Optimization of Materials, solution to industrial problems, solution to environmental problems etc., all leading to sustainable development. Some notable thrust areas under which student projects were carried out are:

- ✓ Composite Materials
- ✓ Construction Material Analysis
- ✓ Pervious Concrete
- ✓ Fiber Reinforced Concrete
- ✓ Geotechnical Studies
- ✓ Air Quality Management
- ✓ Wastewater Treatment.

The details of project and its outcomes are listed below in the table.

Table 3.1 List of Projects in Each Academic Year

Academic Year	Design Projects	Main Projects	No. of. Conference Publications
2016-17	18	19	14
2017-18	18	18	18
2018-19	18	20	17
2019-20	15	18	11
Total	69	75	60

Notable Effectiveness:

Students were motivated to participate in competitions and conferences in reputed institutions including IITs, NITs, and Anna University Campuses and won prizes. Following are some of the achievements of students:

- Received Rs. 50,000/- cash award for an innovative project on “Intelligent Water Estimation for Agricultural Crops” at Sai Ram Engineering College, Chennai
- Dr.Kalam Young Achiever Award from World Youth Federation at MIT, Chennai
- Won First Prize in the event on “Aquanomics” in “CEAFEST’18”,IITM.

- Won First Prize in Techno – Management Symposium in “GYAN MITRA’18”.
- Won First Prize in National Level Technical Symposium “RENAISSANCE’17”.
- Won overall championship in GYAN MITRA’17.

In addition to curriculum based projects students were also involved in various in-house projects as listed below.

Table 3.2 List of In-house projects in each academic year

S.No	Area	Title	Batch
1.	Application Development in Agriculture Applications	Intelligent Water Estimation for Agricultural Crops	2013-2017
2.	Total station surveying	Rotary School Ground Surveying	2014-2018
3.	Developing Model & Total station surveying	RIT Miniature, Leveling Sports arena	2015-2019
4.	Total station surveying	Leveling Sports arena	2016-2020
5.	Total station surveying	P.A.C.R Campus- Landscape Surveying, Marking RIT Sports Arena and Pathway.	2017-2021

Internship Details:

Students were encouraged to go on internships and carry out industry based projects during their final semester. The list of students who did industry based final year projects and the organizations which provided internships are as listed below:

Table 3.3 Internship Details

Academic Year	Name of the company	Number of students
2016-2017	L&T IDPL, Chennai	4
2017-2018	Sobha Limited, Bangalore	4
	Transportation Engineering Division, IIT, Madras	1
2018-2019	Ramco Research & Development Centre, Chennai	2
	Utracon Structural Systems Pvt. Ltd., Chennai	11
	C.R.Narayana Rao (Consultants) Pvt. Ltd., Chennai.	2
	Kartya Constructions Pvt. Ltd. (KVT Concrete), Madurai.	4
	Ramco Industries, Chennai	2
	Pentagon Surveys Pvt. Ltd., Bangalore	5
	Chettinad Group of Companies, Chennai	3
	Exeter Premedia Services Private Ltd.	1
2019-2020	L&T IDPL, Chennai	7
	Utracon Structural Systems Pvt Lt.	3

MOU with Organizations:

MOU's have been signed with the following organisations.

1. Ramco Group of Textile Division, Rajapalayam
2. Ramco Cements Ltd.,Chennai.
3. Ramco systems Ltd.,Chennai.
4. Ramco Industries Ltd., Chennai
5. Infinity PMC Private Limited, Chennai
6. Gowri House Metal Works, Rajapalayam
7. Dynamic TEK, Chennai
8. MEDSBY Healthcare & Engineering Solutions, Coimbatore
9. Crystal Delta
10. TCSion, Chennai.
11. Geons Logix
12. M.S. Chellamuthu Trust,Madurai
13. Enthu Technology Solutions India Private Limited, Coimbatore
14. NI Systems Private Limited, Bangalore.
15. Imarticus Learning Pvt Ltd

Industrial Visit Details:

Table 3.4 Industrial Visit Details

Academic Year	Year	Company Visited	Strength
2016-17	I & II	The Ramco Cements Limited Virudhunagar	241
	III	Vikram Sarabai Space Centre, Kerala	
	IV	Ramco Ready Mix Concrete, IIT, CSIR – Structural Engineering Research Centre, Chennai	
2017-18	I	Rajapalayam Mills Limited, Rajapalayam	237
	II	The Ramco Cements Limited Virudhunagar	
	III	Kerala State Science and Technology Museum	
	IV	Cochin Port Trust &International Airport.	
2018-19	II & III	Neptune Ready mix Concrete Pvt. Ltd., Kochi Port	136
	IV	Bengaluru Metro Rail Corporation Ltd & KSR Dam	
2019-20	I	Rajapalayam Mills Limited	122
	II	Neptune Readymix Concrete Pvt. Ltd., Kochi Port.	
	III	Bangalore Metro and KRS Dam	
	IV	URC Construction and KRS Dam	

Placement Details:

The placement details of various academic years are shown below.

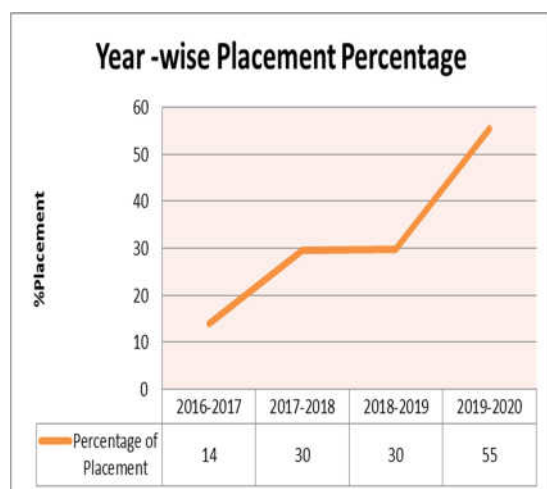


Fig 3.1 Year-wise placement percentage

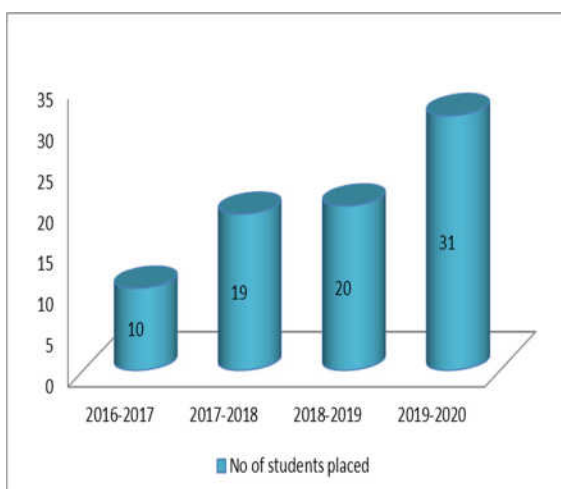


Fig 3.2 Placement Details

Table 3.5 Year-wise placement percentage

Year	Total Strength	Placed Strength	% Placement
2016-2017	71	10	14
2017-2018	64	19	30
2018-2019	67	20	30
2019-2020	56	31	55

Company wise Placement Details (2019-2020):

Table 3.6 Company wise placement details (2019-2020)

Sl. No.	Name of the Company	No. of Students Placed
1.	Ramco Cements	03
2.	Milekal Engineering	01
3.	SMART Training Resources, Chennai	05
4.	ETHNUS, Bangalore	02
5.	Jasper International Engineering Consultants	15
6.	Speed Team, Tenkasi	05
Total		31

HIGHER STUDY DETAILS:

21 students of various batches of Civil Engineering pursued their higher education in reputed Institutions. 4 out of 21 have pursued higher studies in foreign Universities.

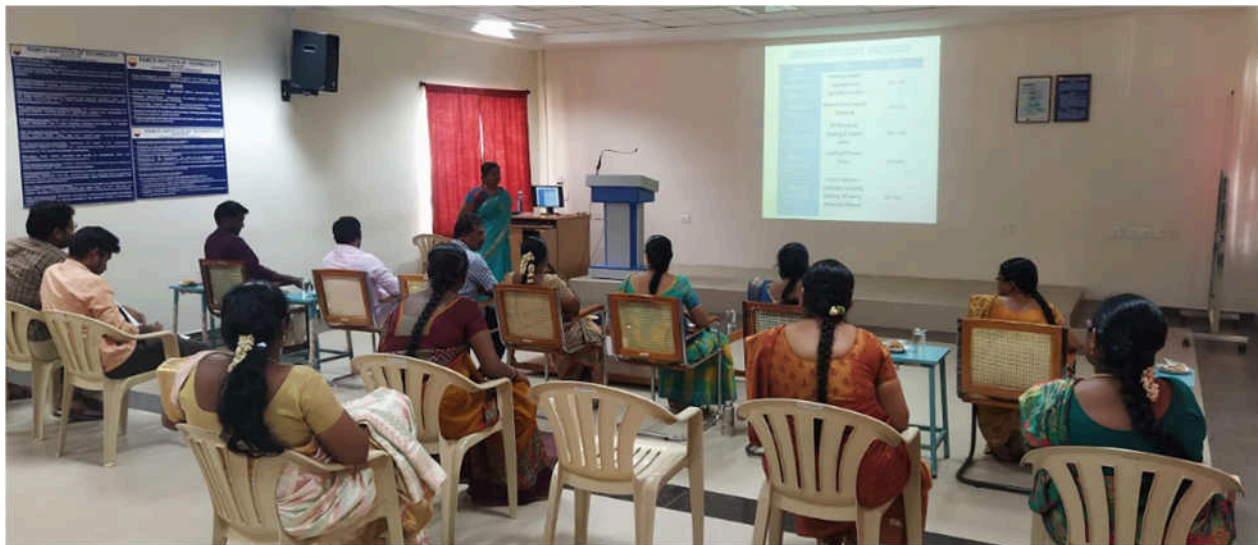


Fig. 3. Program Effectiveness, presented by Mrs. C. Subha AP(SG)/Civil

Remarks from Members:

- All Student projects should be presented in reputed conferences and the papers should be published in international journals.
- All Students should attend In-Plant Trainings and internships to acquire required skills for industry.
- Science Day projects, Design and Fabrication projects and final year projects can be consolidated and classified for the scope of patenting, research studies and product development.
- Members recommended improving the turnout ratio of paper presentation in International Conferences organized by reputed institutions.

4. LIST OF DOCUMENTATION OF ACTIVITIES LEADING TO QUALITY IMPROVEMENT

Presenter: Mrs.D.Darling Helen Lydia, Assistant Professor/Civil

Mrs.D.Darling Helen Lydia presented the improvements and progress in teaching – learning process, Academic research, sponsored research, Consultancy and Development activities.

Teaching Learning process:

- Explained how implementation of innovative teaching to the problematic and theory subjects through the ICT tools like Smart class room. She also explained the use of Online resources and Learning Management System (LMS) usage to improving class room delivery.
- Discussed effective, Transparent Assessment and Evaluation techniques for conducting online quizzes, Scheme of Evaluation and effectiveness of other department experts in project review assessment and consolidated Rubrics for providing better assessment technique in Assignments valuation

Academic, Sponsored Research, Consultancy and Developmental Activities:

- Described the number of faculty pursuing Ph.D., papers published in conferences, ongoing sponsored research projects and their funding agencies.
- Discussed consultancy projects and work carried out in the Department.
- Table 4.1 presents the details of consultancy works

Table 4.1 Details of sponsored research and consultancy works

Academic Year	Name of the client	Amount Received
2019-2020	PWD,Rajapalayam	6,750.00
	Sri Jeyajothi Textiles Mills (P) Ltd.	2,250.00
	Panchayat Union Elementry School,Solapuram	9,000.00
	Total	18,000.00



Fig. 4. List of Documentation of Activities Leading to Quality Improvement, presented by Mrs.D.Darling Helen Lydia, AP/Civil

Remarks from Members:

- All activities leading to quality improvement can be presented in a detailed manner
- Steps need to be taken to increase consultancy work and sponsored research projects by applying to various funding agencies.
- Potential Student projects should be identified and submitted for project proposal.
- All the faculty members were asked to enhance their class room delivery with innovative techniques.
- Committee members recommended using Learning Management System (LMS) platform for all possible courses.

5. REPORT ON PROGRAM ACTIVITIES, PROGRESS AND STATUS

Presenter: Mr.R.Muruganantham, AP/Civil

The effective functioning of professional societies & clubs and the activities organized were presented. The key activities and these effectiveness were presented and the following points were highlighted.

- Around 90 students have registered with ICI Students Chapter and are actively participating in events conducted by RIT- STRUCTA and ICI Students Chapter periodically.
- Academic year wise students participation in symposium/Workshop/Paper presentation was consolidated as listed below:
- Table 5.1 presents the details of activities carried out during the academic year 2019 – 2020 by various professional societies, clubs and association

Table 5.1 Details of clubs, societies and association activities

Sl. No.	Club/Society	Academic Year	No Activities Conducted
1.	RIT - STRUCTA	2019 – 2020	16
2.	ICI – Student Chapter	2019 – 2020	5
3.	RIT – Alumni Association	2019 – 2020	2



Fig.5. Report on program activities, progress and status, presented by Mr.R. Muruganantham, AP/Civil

Remarks from Members:

- RIT – STRUCTA and ICI – Student Chapter Society should perform well in the coming academic year. They also suggested conducting more events in future.
- Committee members suggested conducting funded workshops, seminars, etc.
- More industrialists can be invited to address the students to give awareness on recent technologies in industries.
- Webinars and Guest lectures can be conducted to fill the gap in syllabus and all events can be mapped with POs and PSOs.

6. FACULTY PARTICIPATION, PAPER PRESENTATIONS/PUBLICATIONS, WORKING MODELS AND RESEARCH

Presenter: Mr.J.Ram Prashath, Assistant Professor/Civil

The details of faculty members' participation and research activities were presented. The following keys points were discussed.

- Faculty members are participating in various workshops, seminars, attending conferences, FDTPs, industrial training and STTP in various reputed industries and institutions in all academic year.
- Publications in journals by the faculty members and the list of the journal titles were presented.
- Listed the conference publications (National and International) in all academic year.
- Mentioned the faculty members with Ph.D. and recognized supervisor details. Documents have been prepared regarding faculty members who have registered for Ph.D and details of faculty who have pursuing Ph.D and details of their recognized guide specific domain were presented.
- Consolidated data of faculty participation in various reputed conference publications (National and International) in all academic year was also presented.

Table 6.1 Details of Workshop / FDP / FDTP / STTP / Industrial training participation

Sl. No.	Academic Year	Number of Events Participation	Level
1.	2019-2020	04	International
		17	National
		11	State
		79	Local
2.	2020-2021	10	International
		34	National
		12	State
		16	Local

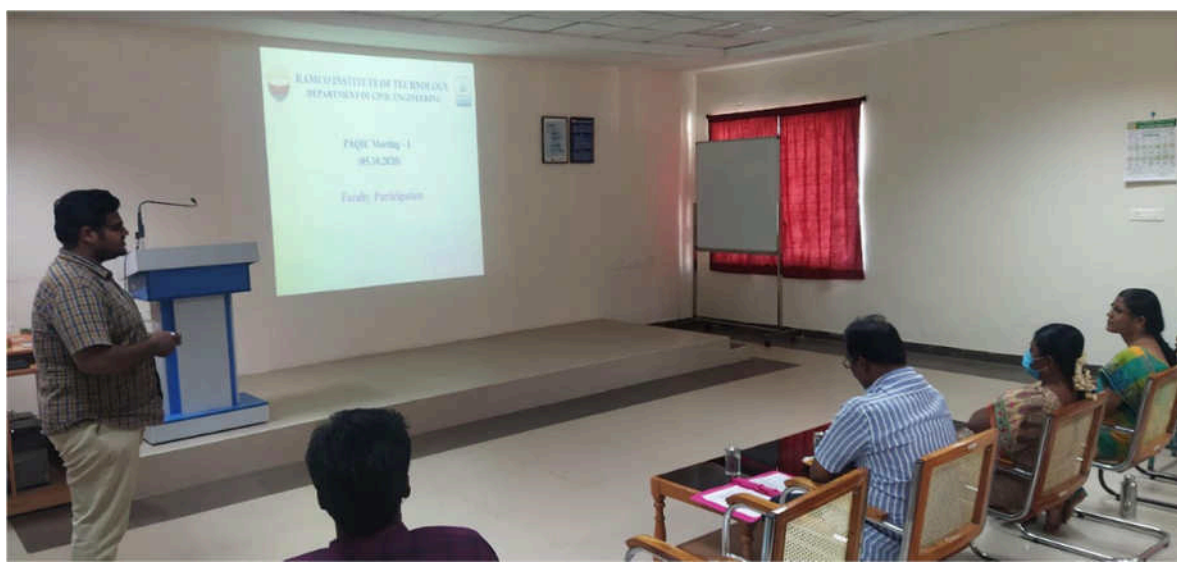
The details of faculty participations in various workshops, FDP, FDTP, STTP, etc. are presented in Table 6.1. Book Chapter publication, contribution as Advisory committee member and external jury member are given in Table 6.2. Details of conference presentation is presented in Table 6.3.

Table 6.2 Details of other contributions

Sl. No.	Academic Year	Contribution	Nos.
1.	2019-2020	External Jury member – Paper Presentation	01
2.	2020-2021	Advisory Committee Member	01
		Book Chapter Publication	01
		Resource Person	02

Table 6.3 Details of conference presentations

Sl. No.	Academic Year	International	National
1.	2019-2020	-	10
2.	2020-2021	12	-



**Fig.6. Report on program activities, progress and status, presented by
Mr. J. Ram Prashath, AP/Civil**

Remarks from Members:

- Faculty member have to be motivated to present research papers in Conference and thereby increase publications in journals too.
- All faculty members have to be encouraged to pursue Ph.D
- Faculty members pursuing can apply for research grants.
- Participation and publication analysis need to be prepared as per NBA requirement.

7. RESULT ANALYSIS

Presenter: Mrs.R.Kalaimani, AP/Civil

The results of 2019 – 2020 Odd semesters and even semesters result and Previous University Rank list students were presented. The following points were highlighted during the presentation.

- The individual subject wise results and scope for improvement in revaluation.
- Civil department results were compared with other branches
- Highlighted the potential for improvement in result which had already been discussed with the students.
- List of prospective students (2016 – 2020 batch) who can secure University Ranks



Fig.7. Result Analysis, Presented by Mrs. R. Kalaimani, AP/Civil

Remarks from Members:

- Committee members suggested comparing the 2019-2020 result with previous year results.
- Committee members observed that results of Mathematics based subjects have declined in II year compared to Theory subjects. Hence, students should be encouraged to solve more problems in tutorial and work on assignments for improving their mathematical skills.
- III year and Final Year students' results are better when compared to other department results. Still some Mathematics based subjects' results are declining. Committee members asked to get feedback from all the students and faculty members for decline in academic results and take necessary steps to improve the results further.

8. ANY OTHER MATTERS

Presenter: Mr.A.Manicka Mamallan, AP/Civil

Future Infrastructure Work:

- RIT Sports Arena
 - Basket ball court
 - Cricket Pitch
 - Landscape Works
 - Storm water drainage
- Preparation of college layout using Total Station.
- Construction of 4th academic block
- Construction of Vinayagar Temple
- Construction of Bust Statue of Founder Chairman

Department Library:

It has been planned to relocate the Department Library with E-Journals and E-Books to the space in front of Soil Mechanics Laboratory since the existing department library is congested.

Model Laboratory:

As per 2017 regulations, a separate model laboratory has been planned for the better understanding of the Irrigation and Environmental Engineering Drawing.

Special Equipment for Consultancy:

It has been decided to purchase soil rig for carrying out standard penetration consultancy work in soil laboratory.

GATE Coaching:

Gate coaching is provided by the GATE FORUM for the students of 2017 – 2021 & 2018 – 2022 batches.

The number of students undergoing Gate Coaching are

2017 – 2021 – 20 students

2018 – 2022 - 07 students

GATE Scorers (2015 – 2019 Batch):

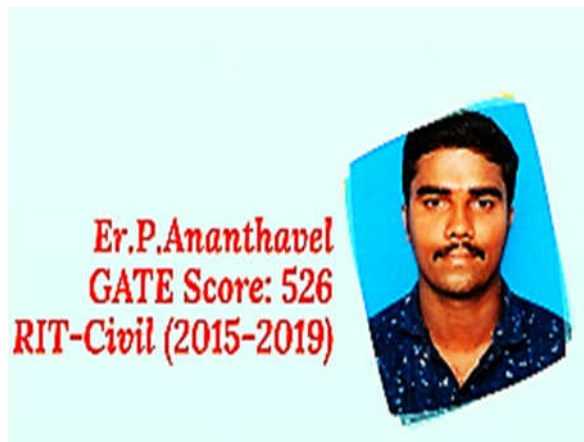


Fig.8.Presentation by Mr. A. Manicka Mamallan, AP/Civil

Remarks from Members:

- Remuneration or Certificate of appreciation to be given to the Students who are involved with Civil work which are carried out in our campus.
- Equipment that is available in the Laboratory beyond the Anna University requirements needs to be listed.

Overall Remarks from Members:

- Steps need to be taken to improve admission, academic result and placement in the coming year.
- The department can discuss with other departments for additional inputs to enhance their PAQIC practices.

Prepared by
(Mr.G.Karthikeyan, AP(SG)/Civil)

Reviewed by
(HOD/Civil)

Approved by
Principal